M Akash Krishnan

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Education

Dr. M.G.R. Educational and Research Institute — Chennai, In

B. Tech in Computer Science - Artificial Intelligence and Data Science - CGPA : 8.5 — *Oct. 2022 - May 2026*

Technical Skills

Languages: Python, MySQL, JavaScript, HTML/CSS, Java
Frameworks: React, Django Framework, Selenium, Streamlit, Node.js, Flask, Express.js, Three.js, FastAPI
Developer Tools: Git, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, AWS Cloud Services, Azure, PowerBI
Libraries: pandas, NumPy, Matplotlib, Scipy, Seaborn, Keras, NLTK, Spacy, Scikit-Learn

Projects

Career Path Recommender | *Python, Flask, React, API* — Aug 2025 – Present

- Developed a full-stack web application using with Flask serving a REST API with React as the front-end
- Engineered a Machine learning model using BERT to recommend suitable career paths based on user interests
- The front-end uses Reddit's Newsfeed that focuses on Career Growth / Career Development posts from several popular Reddit pages.
- The BERT Model is fine-tuned for semantic similarity to match student preferences with relevant career streams
- The Dedicated Output consisting of the Career path the model suggests is exported in a .pdf format, the Graph can be exported Vertically / Horizontally.

Football Deep Learning Model | *PyTorch, XGBoost, Scikit-Learn, Streamlit* — Nov 2025 – Present

- The model prioritized sensitivity, achieving a high Recall of 88.89% to ensure the winner is captured, with a trade-off resulting in 3.37% Precision and an F1 Score of 6.50%.
- To mimic voting bias, you implemented a custom feature weighted at 40% for League Goals, 25% for UCL Goals, and 35% for Model Probability
- The solution deploys specialized Deep Learning architectures, alongside XGBoost, to process football statistics from 2005 to 2025.
- The project runs locally on an RTX 4050 (no cloud GPU) with the specific target of predicting the 2026 Ballon d'Or and Champions League winners.
- To account for "Tournament DNA" in the UCL model, you implemented a hard-coded 0.15 probability boost for legacy clubs (Real Madrid, Bayern, Liverpool), ensuring the predictions respect historical dominance alongside raw stats.

Resume-Analyzer | *PyTorch, Scikit-Learn, API, Streamlit* — Nov 2025 – Present

- Designed a privacy-first application that decouples inference logic, using Local Embeddings (all-MiniLM-L6-v2) for sensitive scoring and offloading complex reasoning to the Zephyr-7B LLM via the Hugging Face API.
- Engineered a vector-based scoring system that converts resumes and JDs into high-dimensional embeddings, utilizing Cosine Similarity to calculate precise "Job Fit" percentages rather than simple keyword counting.
- Developed a "Smart Pre-Screening" feature where the AI autonomously detects skill gaps and generates context-aware Yes/No interview questions, simulating a real-time recruiter phone screen.
- Built a custom text extraction pipeline using pdfplumber and RegEx to handle complex multi-column PDF layouts, ensuring high-fidelity data cleaning before vectorization or LLM processing.
- Deployed a modern, responsive UI using Streamlit and Custom CSS, featuring real-time metric cards and an interactive feedback loop that helps candidates optimize their profiles for Applicant Tracking Systems (ATS).

Certifications

IBM: Python, Java, Business Intelligence, Cloud Platform
NPTEL: Python for Data Science
Coursera: Django Framework, API, Version control